Tongren Xiao

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Research Interests

Financial volatility modeling and forecasting; shape-space methods and shape-based representations for financial time series; forecast calibration and underestimation-risk control; Schubert calculus and algebraic geometry; selected difficult mathematical conjectures.

Education

Sun Yat-sen University, School of Mathematics 2023–September 2027 (expected)
Undergraduate Student in Mathematics
GPA: 3.00/4.00; average mark: 80/100

Research Experience

Research in Quantitative Finance 2025–present
Advisor: Associate Professor Yirong Huang, Sun Yat-sen University
Research on financial volatility modeling and forecasting, including hybrid LSTM–GARCH comparison, geometric analog forecasting of realized volatility, and post-processing calibration for forecast underestimation risk.

Undergraduate Research in Schubert Calculus 2025–present
Advisor: Professor Changzheng Li, Sun Yat-sen University
Research on positivity properties of quantum double Schubert polynomials and related structures in Schubert calculus.

Research on Difficult Mathematical Conjectures 2026–present
Developed rigorous partial results and explicit remaining-gap analyses for the three-frame submatrix problem in matrix analysis and for the Erdős–Straus conjecture through fixed-shift obstructions, large-sieve estimates, and an optimization of Dahan's sieve constants.

World Models for Financial Applications 2026–present
Anchor-Conditioned Probabilistic World Models for Multi-Asset Realized-Volatility Paths (work in progress). Exploring latent state-space dynamics, HAR–IV anchoring, multi-step residual likelihood, and joint path realism for financial volatility forecasting.

Selected Manuscripts and Research Outputs

- **Shape Is Useful: Geometric Analog Forecasting of Realized Volatility.** Tongren Xiao and Shuhang Chen. Manuscript, 2026.
- **A Comparative Study of Hybrid LSTM Frameworks for Volatility Forecasting in the NASDAQ-100 and S&P 500 Markets.** Yue Yihua, Rong Jia, Lv Zimeng, Xiao Tongren, and Li Ke. Manuscript, 2025.
- **Graham Positivity of Quantum Double Schubert Polynomials.** Tongren Xiao and Yihua Yue. Manuscript, 2026.
- **On the Submatrices with the Best-Bounded Inverses for Matrices with Three Orthonormal Columns.** Research note, 2026.
- **Fixed-Shift Obstructions and Optimized Sieve Constants for the Erdős–Straus Conjecture.** Research note, 2026.

Coursework and Academic Teaching

Selected advanced coursework includes two graduate-level courses, one honors course, and one interdisciplinary course: *Selected Readings of Academic Articles*, *Harmonic Analysis on Groups*, *Algebraic Topology*, and *Stochastic Processes*. In *Selected Readings of Academic Articles*, led presentations on Chapters 1 and 3 of Emily Riehl's *Category Theory in Context*.

International Academic Training

Introduction to Option Pricing 7 March–25 May 2025
International research-based learning course taught by Johannes Ruf; grade: 81.50.

Honors and Projects

- Honorable Mention, 2026 COMAP Mathematical Contest in Modeling (MCM), Problem B.
- Guangdong Provincial Second Prize, 2025 Contemporary Undergraduate Mathematical Contest in Modeling (CUMCM), Problem A.
- Provincial Undergraduate Innovation and Entrepreneurship Training Program, Project No. 20251647: *Asset Volatility Modeling Based on Multi-Option Portfolio Strategies*.

Technical Skills

LaTeX for academic writing and manuscript preparation; MATLAB for mathematical modeling and numerical computation; mathematical proof and technical writing; linear algebra, matrix analysis, and abstract algebraic structures.